PROBLEM8:

PROBLEM STATEMENT:

Manufacturing Cost and Profitability Analysis: Calculate material costs, labour charges, wastage, machine costs, and selling prices; identify whether production meets a specified profit target.

OBJECTIVE:

To develop a Python program to calculate the total manufacturing cost by considering material costs, labour charges, wastage, and machine costs. Calculate the selling price and profit, then determine whether production meets the specified profit target.

PROGRAM:

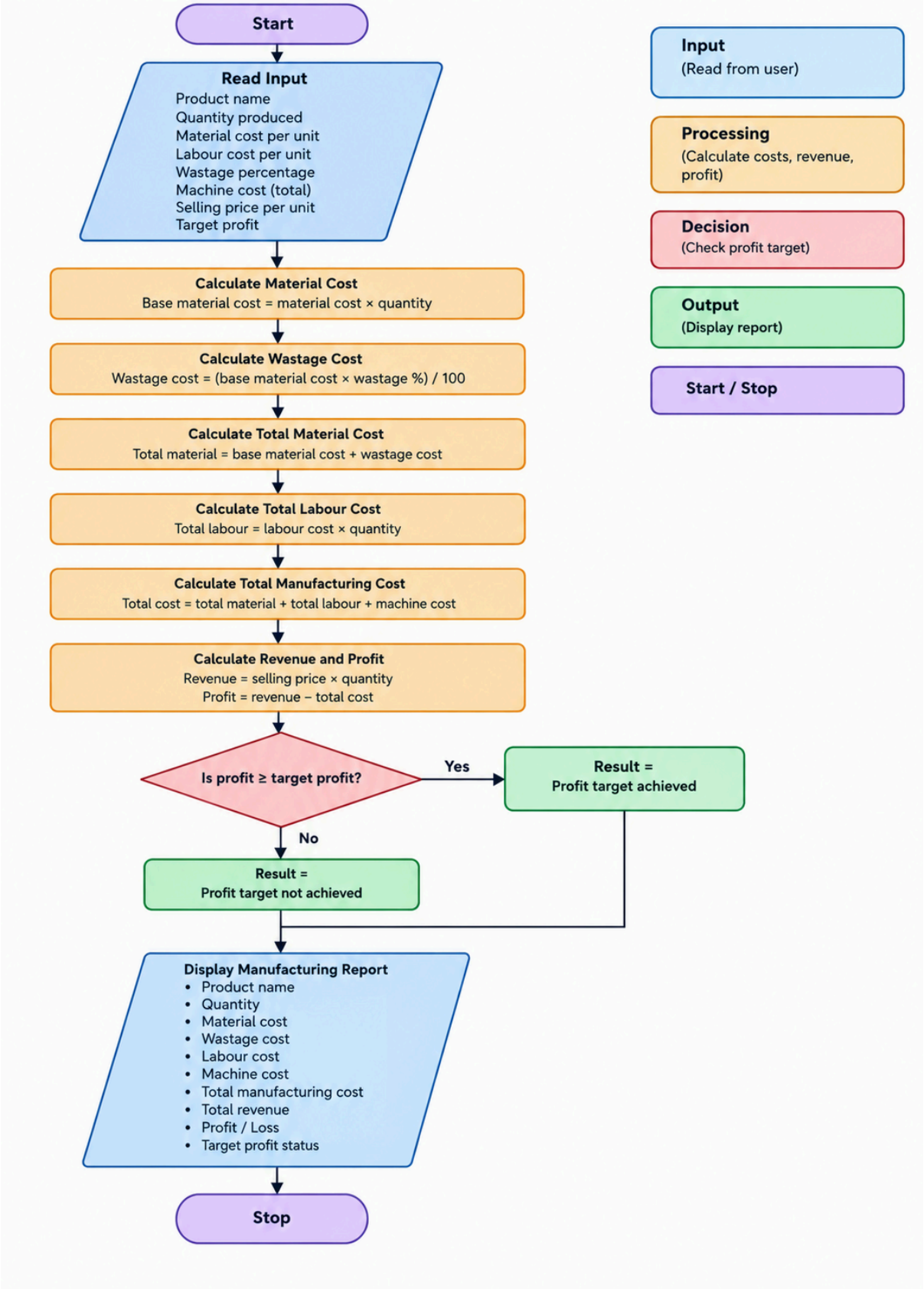
```
# Manufacturing Cost and Profitability Analysisproduct = input("Enter product name: ")quantity = int(input("Enter production quantity: "))material_cost = float(input("Enter material cost per unit: "))labour_cost = float(input("Enter labour cost per unit: "))wastage_percent = float(input("Enter wastage percentage: "))machine_cost = float(input("Enter total machine cost: "))selling_price = float(input("Enter selling price per unit: "))profit_target = float(input("Enter required profit percentage: "))# Calculate material costtotal_material = material_cost * quantity# Calculate wastage costwastage_cost = total_material * wastage_percent / 100# Calculate total labour costtotal_labour = labour_cost * quantity# Calculate total manufacturing costtotal_cost = total_material + wastage_cost + total_labour + machine_cost# Calculate revenue and profitrevenue = selling_price * quantityprofit = revenue - total_cost# Calculate profit percentageif total_cost > 0:    profit_percent = (profit / total_cost) * 100else:    profit_percent = 0# Display resultsprint("\n----- MANUFACTURING REPORT -----")print("Product:", product)print("Total Material Cost: Rs.", round(total_material, 2))print("Wastage Cost: Rs.", round(wastage_cost, 2))print("Total Labour Cost: Rs.", round(total_labour, 2))print("Machine Cost: Rs.", round(machine_cost, 2))print("Total Manufacturing Cost: Rs.", round(total_cost, 2))print("Total Revenue: Rs.", round(revenue, 2))print("Profit/Loss: Rs.", round(profit, 2))print("Profit Percentage:", round(profit_percent, 2), "%")if profit_percent >= profit_target:    print("Result: Profit target achieved.")else:    print("Result: Profit target not achieved.")print("--------------------------------")
```

ALGORITHM:

1. Start.
2. Input product name and production quantity.
3. Input material cost, labour cost, wastage percentage, machine cost, selling price, and required profit percentage.

4. Calculate total material cost = material cost per unit × quantity.
5. Calculate wastage cost = total material cost × wastage percentage / 100.
6. Calculate total labour cost = labour cost per unit × quantity.
7. Calculate total manufacturing cost by adding material, wastage, labour, and machine costs.
8. Calculate total revenue = selling price per unit × quantity.
9. Calculate profit = total revenue − total manufacturing cost.
10. Calculate profit percentage = profit / total manufacturing cost × 100.
11. Compare the profit percentage with the required profit target.
12. Display all costs, revenue, profit, and the target result.
13. Stop.

FLOWCHART:


Start
Read Input
Product name
Quantity produced
Material cost per unit
Labour cost per unit
Wastage percentage
Machine cost (total)
Selling price per unit
Target profit
Calculate Material Cost
Base material cost = material cost × quantity
Calculate Wastage Cost
Wastage cost = (base material cost × wastage %) / 100
Calculate Total Material Cost
Total material = base material cost + wastage cost
Calculate Total Labour Cost
Total labour = labour cost × quantity
Calculate Total Manufacturing Cost
Total cost = total material + total labour + machine cost
Calculate Revenue and Profit
Revenue = selling price × quantity
Profit = revenue – total cost
Is profit ≥ target profit?
Yes
Result = Profit target achieved
No
Result = Profit target not achieved
Display Manufacturing Report
Product name
Quantity
Material cost
Wastage cost
Labour cost
Machine cost
Total manufacturing cost
Total revenue
Profit / Loss
Target profit status
Stop
Input
(Read from user)
Processing
(Calculate costs, revenue, profit)
Decision
(Check profit target)
Output
(Display report)
Start / Stop

PSEUDOCODE:

```
START

INPUT product, quantity
INPUT material_cost, labour_cost
INPUT wastage_percent, machine_cost
INPUT selling_price, profit_target

total_material = material_cost * quantity
wastage_cost = total_material * wastage_percent / 100
total_labour = labour_cost * quantity

total_cost = total_material + wastage_cost
           + total_labour + machine_cost

revenue = selling_price * quantity
profit = revenue - total_cost

IF total_cost > 0:
   profit_percent = profit / total_cost * 100
ELSE:
   profit_percent = 0

DISPLAY total_material
DISPLAY wastage_cost
DISPLAY total_labour
DISPLAY machine_cost
DISPLAY total_cost
DISPLAY revenue
DISPLAY profit
DISPLAY profit_percent

IF profit_percent >= profit_target:
   DISPLAY "Profit target achieved"
ELSE:
   DISPLAY "Profit target not achieved"

STOP
```

OUTPUT:

Assume the following input values:

```
Enter product name: Chair
Enter production quantity: 100
Enter material cost per unit: 200
```

Enter labour cost per unit: 50
Enter wastage percentage: 10
Enter total machine cost: 2000
Enter selling price per unit: 350
Enter required profit percentage: 20

Output:

Product: Chair
Total Material Cost: Rs. 20000.0
Wastage Cost: Rs. 2000.0
Total Labour Cost: Rs. 5000.0
Machine Cost: Rs. 2000.0
Total Manufacturing Cost: Rs. 29000.0
Total Revenue: Rs. 35000.0
Profit/Loss: Rs. 6000.0
Profit Percentage: 20.69 %
Result: Profit target achieved.